AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

ULTRA VIOLET (UV) DISINFECTION SYSTEM

SAFE, CHEMICAL FREE & ENVIRONMENT FRIENDLY DISINFECTION

FOR WATER & WASTEWATER APPLICATIONS

- Effective disinfection (Bacteria, Virus, Protozoa)
- No chemicals & no by-products
- Low operating cost
- Automatic operation with alarms
- SS-304 / SS-316 construction
- Wide range from 1 to 100 m³/hr
- Compact, skid mounted design
- Suitable for continuous operation



UV SYSTEM 1 – 10 m³/hr (SS-304 / SS-316)



UV SYSTEM 10 – 50 m³/hr (SS-304 / SS-316)

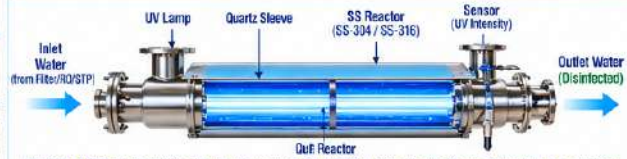


UV SYSTEM 50 – 100 m³/hr (SS-304 / SS-316)



CONTROL PANEL With UV Intensity & Alarms

WORKING PRINCIPLE OF UV DISINFECTION



UV light at 254 nm penetrates the microorganism DNA/RNA and inactivates them, preventing reproduction. It effectively disinfects bacteria, viruses, protozoa and other pathogens without use of chemicals.

TYPICAL APPLICATIONS



Drinking Water Plants



RO Permeate Disinfection



STP Tertiary Treatment



ETP Reuse & Recycling



Hotels & Resorts



Hospitals & Healthcare



Food & Beverage Industry



Pharmaceutical Industry

KEY FEATURES

- High disinfection efficiency (up to 99.99%)
- Chemical free and no residuals
- Compact and modular design
- SS-304 / SS-316 construction
- Low maintenance and operating cost
- UV intensity sensor with alarm
- Lamp life indicator and hour meter
- Automatic on/off operation
- Suitable for continuous operation
- Wide range from 1 to 100 m³/hr
- Skid mounted or inline type
- Suitable for potable water, RO, STP, ETP and industrial applications

TYPICAL INLET & OUTLET PARAMETERS

Parameter	Unit	Typical Inlet	Typical Outlet (after UV)
pH	-	6.5 – 8.5	6.5 – 8.5
Turbidity	NTU	< 5	< 1
TSS	mg/L	< 10	< 5
Colour	Hazen	< 10	< 5
BOD	mg/L	< 10	< 10
COD	mg/L	< 50	< 50
Total Coliform	MPN/100 mL	$10^3 – 10^6$	< 2
E. coli	MPN/100 mL	$10^3 – 10^6$	< 2

Note: For best performance, water shall be low in turbidity and free from colour. UV is used for disinfection and does not remove TDS, BOD or COD.

MATERIAL OF CONSTRUCTION (MOC)

Item	Material
UV Reactor Body	SS-304 / SS-316
Inlet / Outlet Flanges	SS-304 / SS-316
Quartz Sleeve	High Purity Quartz
UV Lamp	Low Pressure High Output (LPHO)
Seals / O-rings	EPDM / Viton
Lamp Housing	SS-304 / SS-316
Control Panel Enclosure	MS Powder Coated / SS-304 (Optional)
Cables & Connectors	Waterproof, IP-67
UV Intensity Sensor	UV Sensor with 4-20 mA (Optional)
Flow Switch	SS / PP (Optional)

TECHNICAL SPECIFICATIONS – UV DISINFECTION SYSTEM (1 m³/hr to 100 m³/hr)

Model	Flow Capacity (m³/hr)	No. of UV Lamps	Lamp Power (Watt)	Total Power (kW)	Inlet / Outlet Size (NB)	Chamber Size (mm) L × Dia (Indicative)	MOC (SS-304 / SS-316)
AQU-UV-1	1	1	30	0.05	25	600 × 89	SS-304 / SS-316
AQU-UV-3	3	1	55	0.08	25	700 × 114	SS-304 / SS-316
AQU-UV-5	5	1	75	0.10	40	900 × 114	SS-304 / SS-316
AQU-UV-10	10	2	75	0.18	50	900 × 168	SS-304 / SS-316
AQU-UV-15	15	3	75	0.27	65	1,000 × 219	SS-304 / SS-316
AQU-UV-20	20	4	75	0.36	80	1,200 × 219	SS-304 / SS-316
AQU-UV-30	30	6	75	0.54	100	1,400 × 273	SS-304 / SS-316
AQU-UV-50	50	8	75	0.72	150	1,600 × 323	SS-304 / SS-316
AQU-UV-75	75	10	75	0.90	200	1,800 × 406	SS-304 / SS-316
AQU-UV-100	100	12	75	1.08	250	2,000 × 406	SS-304 / SS-316

Note: Above specifications are indicative. Actual size and configuration may vary based on water quality and UV dose requirement.

TYPICAL BILL OF MATERIAL (BOM)

Sr. No.	Description	Qty.	MOC
1	UV Reactor Chamber	1	SS-304 / SS-316
2	UV Lamp (LPHO)	As per capacity	-
3	Quartz Sleeve	As per lamps	High Purity Quartz
4	Ballast / Electronic Driver	As per lamps	-
5	UV Intensity Sensor (Optional)	1	-
6	Flow Switch (Optional)	1	SS / PP
7	Control Panel with Hour Meter, Alarms & MCB	1	MS / SS-304 (Optional)
8	Inlet / Outlet Flanges	1 Set	SS-304 / SS-316
9	Seals / O-rings	1 Set	EPDM / Viton
10	Cables, Connectors & Accessories	1 Set	-
11	Skid / Support	1	SS-304 / MS (Optional)

OPTIONAL ACCESSORIES



UV Intensity Sensor



Flow Switch



Digital UV Controller



SS Skid with Panel

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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CLEANER WATER GREENER TOMORROW

1. CIVIL WORK IN PROGRESS – STP + TERTIARY TREATMENT (MGF, ACF, UV) – 100 KLD



2. MGF, ACF, SOFTENER & UV INSTALLATION – 100 KLD (TREATED WATER)



3. COMPLETE STP PACKAGE PLANT WITH UV – 50 KLD (MSRL / MSEP)



4. MGF, ACF, SOFTENER, UV – 30 KLD (COMPACT SKID MOUNTED PLANT)



5. UV DISINFECTION SYSTEM – INSTALLATION IN PROGRESS (TREATED WATER LINE – 20 KLD)



6. COMPLETE PLANT – STP + MGF + ACF + SOFTENER + UV – 100 KLD (CIVIL + ELECTRO-MECHANICAL)

